

Agentic Earning Signal Intelligence System (AESIS)

Problem Statement

During peak earnings season, a mid-sized investment firm receives **50 or more earnings call transcripts in a single week**, each between 8,000 and 15,000 words. Analysts are simultaneously managing positions and serving clients, leaving the most actionable content unread or processed too late to trade on. The highest-value signal in these calls is rarely the headline numbers; it is **how executives communicate**. Shifts in tone, rising hedging language, and evasive Q&A responses carry measurable predictive value for post-earnings price movement. Most firms have no systematic means of detecting these patterns before human analysts can even open the document.

Proposed AI Solution

AESIS is a **text-based, agentic AI system** that functions as an autonomous first responder to financial disclosures. The moment a new transcript is released, AESIS ingests the document, retrieves up to 12 quarters of historical management language for that company, and runs a structured Divergence Analysis, comparing what executives are saying now against prior commitments. The system outputs a **Signal Score**: a weighted metric that flags when executive tone conflicts with reported financial data, delivered automatically, before any analyst prompt is required. Unlike traditional search-based tools, AESIS does not wait to be asked, it acts.

Business Value

Prototype backtesting demonstrates an **80% reduction in per-transcript processing time** (from ~30 minutes to ~5 minutes) and a directional accuracy of approximately **60% for post-earnings price movement** prediction versus a 50% random baseline. Simulated portfolio testing against a \$1M benchmark yielded a +0.5% return improvement. Scaled across a 500-company coverage universe and multiple analysts, these efficiency and alpha-generation gains compound materially each earnings season.

Key Findings

Academic validation is strong: research in the **Journal of Finance** and the **Review of Financial Studies** confirms that linguistic patterns in earnings calls are statistically associated with post-earnings returns. Market investment confirms commercial appetite: AlphaSense raised over \$650M and Kensho Technologies was acquired by S&P Global for \$550M on the strength of NLP-driven transcript intelligence. Data sourcing is fully feasible using SEC EDGAR, Yahoo Finance, and public datasets. A prototype pipeline successfully processed 10–15 transcripts per week autonomously, confirming technical readiness. The principal operational risk — analyst adoption — is mitigated through an explainability layer that traces every signal to the specific source sentences driving it, paired with a human-in-the-loop validation step that preserves analyst authority.

Implementation Path

The rollout follows a three-phase roadmap:

- (1) **Ingestion & Infrastructure** - data pipeline, SEC EDGAR integration, and preprocessing (Months 1–2)
- (2) **Signal Engine** - divergence analysis, Signal Score generation, and backtesting (Months 3–5)
- (3) **Integration & Deployment** - analyst workflow integration and stakeholder rollout (Months 6–8)

Primary KPI's	Analyst processing time ≤ 5 min / transcript at full deployment
Secondary KPI's	Directional accuracy ≥ 55%, False-positive rate < 25%

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